

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 1 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	03-08-2026	Duration:	60 minutes

Code of Conduct

I Jasim Asmaan A (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Jasim Asmaan A

Question

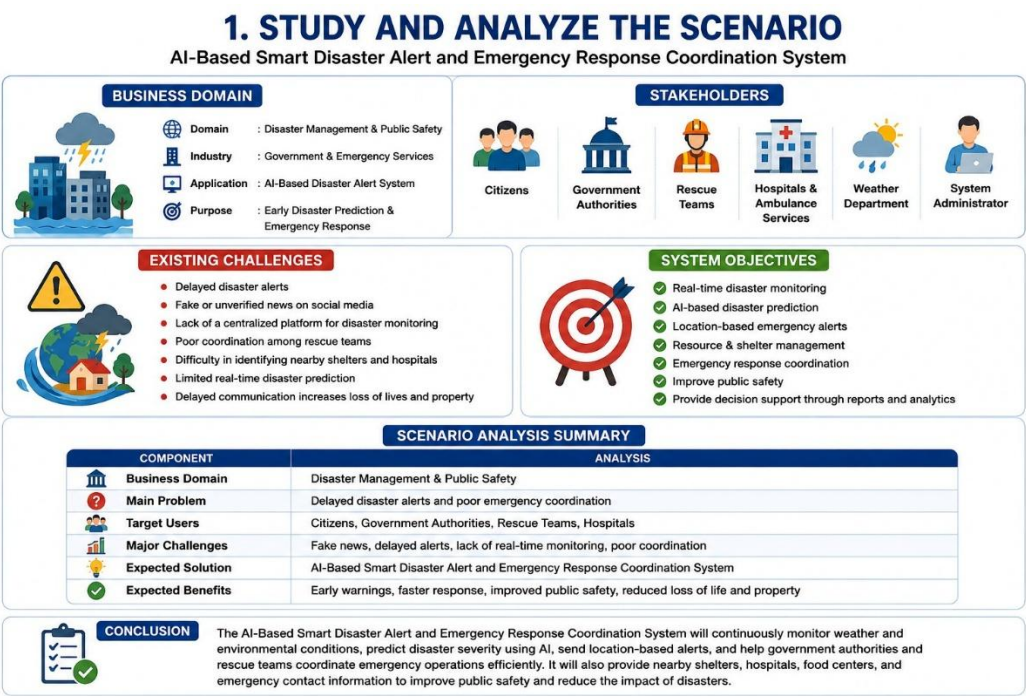
Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1.Study and Analyse the Scenario:

1.1Carefully understand the given problem statement.

Problem Statement

The **AI-Based Smart Disaster Alert and Emergency Response Coordination System** is a disaster management application designed to provide early warnings and improve emergency response during natural disasters such as floods, cyclones, earthquakes, landslides, and heavy rainfall. Existing disaster alert systems mainly depend on television, radio, SMS, and social media, which may delay important information and sometimes spread unverified news. This project aims to develop an intelligent system that continuously monitors real-time weather and environmental data, predicts disaster severity using Artificial Intelligence, sends location-based alerts to citizens, and helps government authorities and rescue teams coordinate emergency operations efficiently. The system also provides nearby shelter locations, hospitals, first-aid centers, food distribution points, and emergency contact information to improve public safety and reduce the impact of disasters.



1.2 Identify the business domain, stakeholders, existing challenges, and system objectives.

DOMAIN:

Aspect	Details
Domain	Disaster Management and Public Safety
Industry	Government, Emergency Services, Smart City Solutions
Application Type	AI-Based Disaster Alert and Emergency Response System
Primary Purpose	Predict disasters, send real-time alerts, and coordinate emergency response
End Users	Citizens, Government Authorities, Rescue Teams, Hospitals

Stakeholders:

Stakeholder	Role in the System
Citizens	Receive disaster alerts and follow safety instructions.
Bank Administrators	Monitor disasters and coordinate emergency response.
Government Authorities	Monitor disasters and coordinate emergency response.
Rescue Teams	Rescue affected people and provide emergency support.
Hospitals & Ambulance Services	Provide medical assistance during emergencies.
System Administrator	Maintains the system, users, and database.

Existing Challenges

- Disaster warnings are sometimes delayed.
- Fake or unverified information spreads through social media.
- Lack of a centralized disaster management platform.
- Emergency resources are difficult to coordinate quickly.
- People may not know the nearest shelters or hospitals.
- Delayed communication increases the impact of disasters.

System Objectives

Objective	Description
Real-Time Monitoring	Monitor weather and environmental conditions continuously.
AI Prediction	Predict disaster severity using AI and real-time data.
Location-Based Alerts	Send alerts only to users in affected areas.
Alert Generation	Help authorities and rescue teams coordinate relief operations.
Emergency Coordination	Prevent unauthorized transactions before completion.
Resource Management	Provide information about nearby shelters, hospitals, food, and first-aid centers.
Public Safety	Reduce disaster impact and protect lives through early warning.

Scenario Analysis Summary

Component	Analysis
Business Domain	Disaster Management and Public Safety
Main Problem	Delayed disaster alerts and poor emergency coordination
Target Users	Citizens, Government Authorities, Rescue Teams, Hospitals
Major Challenges	Delayed alerts, fake news, lack of real-time monitoring, poor coordination
Expected Solution	An AI-based disaster alert system that predicts disasters and provides real-time emergency response support
Expected Benefits	Early disaster warnings, faster rescue operations, improved public safety, and reduced loss of lives and property

2. Identify Functional and Non-Functional Requirements

2.1 List all functional requirements required by the proposed system.

Requirement ID	Functional Requirement	Description
FR1	User Registration and Login	Allow citizens, administrators, and rescue teams to securely register and log in using authenticated credentials.
FR2	User Profile Management	Enable users to update personal information and location details.
FR3	Transaction Processing	Continuously monitor weather conditions and environmental data using APIs and sensors.
FR4	AI-Based Disaster Prediction	Send instant disaster alerts to users based on their current location
FR5	Location-Based Alert System	Display nearby shelters, hospitals, food centers, and first-aid facilities.
FR6	Shelter and Resource Finder	Provide emergency helpline numbers and government support information.
FR7	Emergency Contact Service	Automatically notify administrators when suspicious transactions are detected.
FR8	Rescue Team Coordination	Allow rescue teams to receive disaster information and manage rescue operations.
FR9	Disaster Report Management	Maintain disaster history and incident reports for future reference..
FR10	Dashboard and Reports	Display disaster statistics, prediction reports, and emergency response details..
FR11	Search and Filter	Search disasters, shelters, emergency contacts, and

		reports based on location or disaster type..
FR12	Notification System	Send alerts through SMS, email, and mobile application notifications.
FR13	Role-Based Access Control	Provide different access permissions for citizens, administrators, and rescue teams.
FR14	Data Backup and Recovery	Automatically back up disaster records and user information to prevent data loss.
FR15	System Administration	Allow administrators to manage users, alerts, resources, and overall system operations.

2.2 Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability

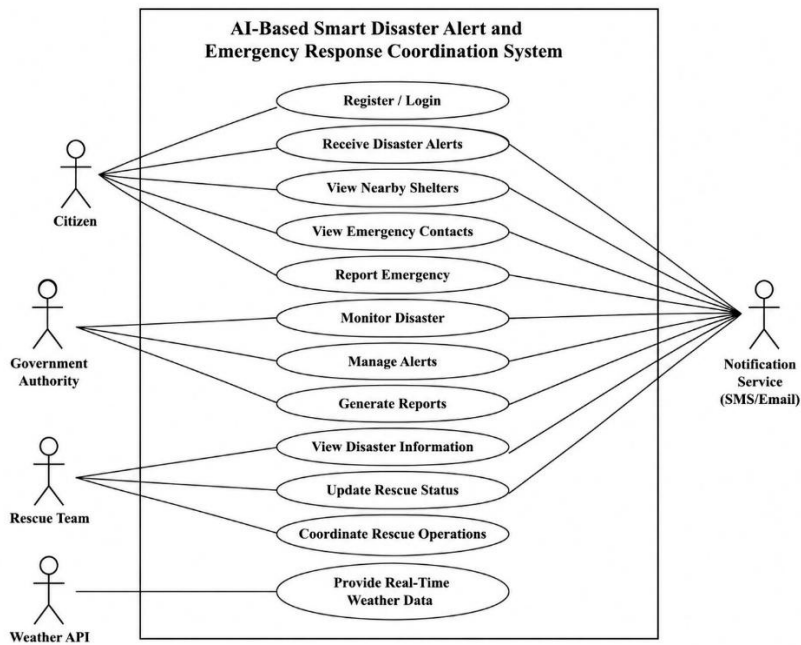
- **Performance:** The system should process real-time weather data quickly and deliver disaster alerts without delay.
- **Security:** User information and disaster data should be protected using secure authentication, authorization, and data encryption.
- **Reliability:** The system should provide accurate disaster predictions and continue functioning during emergency situations.
- **Usability:** The application should have a simple, responsive, and user-friendly interface that is easy for everyone to use.
- **Scalability:** The system should support a growing number of users, disaster records, and emergency resources without affecting performance.
- **Maintainability:** The software should have a modular design so that new features and updates can be added easily.
- **Availability:** The system should be available 24×7 to ensure users can receive emergency alerts and access disaster information at any time..

3. Identify Actors and Use Cases

3.1 Identify all primary and secondary actors interacting with the system.

Actor	Type	Role
Citizen	Primary	Receives disaster alerts, views emergency information, and requests assistance.
Government Authority	Primary	Monitors disaster situations, manages alerts, and coordinates emergency response.
Rescue Team	Primary	Responds to emergencies, rescues affected people, and updates rescue status.
Weather API / IoT Sensors	Secondary	Provides real-time weather and environmental data for disaster prediction.
Notification Service (SMS/Email)	Secondary	Sends disaster alerts and emergency notifications to users.
Actor	Major Use Cases	Actor
Citizen	Register/Login, Receive Disaster Alerts, View Nearby Shelters, View Emergency Contacts	Citizen
Government Authority	Monitor Disasters, Manage Alerts, Allocate Resources, Generate Reports	Government Authority
Rescue Team	View Disaster Information, Rescue Victims, Update Rescue Status	Rescue Team
Weather API / IoT Sensors	Provide Weather Data, Send Environmental Data	Weather API / IoT Sensors
Notification Service	Send SMS, Email, and Mobile App Notifications	Notification Service

3.3 Draw a Use Case Diagram representing the interactions between actors and the system.



4 Prepare the Software Requirement Specification (SRS)

4.1 Introduction

The **AI-Based Smart Disaster Alert and Emergency Response Coordination System** is a smart disaster management application developed to provide early disaster warnings and improve emergency response during natural disasters such as floods, cyclones, earthquakes, and landslides. The system uses Artificial Intelligence to analyze real-time weather and environmental data, predict disaster severity, and send location-based alerts to affected users. It also helps government authorities and rescue teams coordinate emergency operations while providing citizens with nearby shelters, hospitals, food centers, and emergency contact information.

4.2 Problem Description

Existing disaster alert systems mainly rely on television, radio, SMS, and social media to share disaster information. These methods may delay important alerts and sometimes spread fake or unverified news, creating confusion among the public. There is also a lack of a centralized system that can predict disaster severity and provide real-time emergency support. This project addresses these issues by offering AI-based prediction, instant alerts, and effective coordination between citizens and emergency authorities.

4.3 Scope of the System

- Monitor real-time weather and environmental conditions.
- Predict disaster type and severity using Artificial Intelligence.
- Send location-based disaster alerts to affected users.
- Display nearby shelters, hospitals, food centers, and first-aid facilities.
- Provide emergency contact numbers and government support information.

- Help government authorities and rescue teams coordinate rescue operations.
- Generate disaster reports and maintain historical records.

4.4 Functional Requirements

Requirement ID	Functional Requirement
FR1	User Registration and Login
FR2	Disaster Monitoring
FR3	AI Risk Prediction
FR4	Alert Notification
FR5	Resource Finder
FR6	Emergency Contacts

4.5 Non-Functional Requirements

Category	Requirement
Performance	Fast processing of real-time disaster data and instant alert notifications.
Security	Secure user authentication and protection of disaster-related information.
Reliability	Accurate AI-based disaster prediction and continuous system operation.
Usability	Simple and user-friendly interface for citizens and emergency authorities.
Scalability	Support an increasing number of users, disaster records, and emergency resources.
Availability	24×7 system availability for continuous disaster monitoring and emergency response.

4.6 Use Case Descriptions

Actor	Use Cases	Actor
Citizen	Register/Login, Receive Alerts, View Nearby Shelters, View Emergency Contacts	Citizen
Government Authority	Monitor Disasters, Manage Alerts, Generate Reports	Government Authority
Rescue Team	View Disaster Information, Update Rescue Status, Coordinate Rescue Operations	Rescue Team

4.7 Data Requirements

Data Type	Description	Data Type
User Data	Name, Email, Mobile Number, Password, Location	User Data
Disaster Data	Disaster Type, Severity Level, Date, Time, Location	Disaster Data
Weather Data	Temperature, Rainfall, Wind Speed, Humidity	Weather Data
Emergency Resources	Shelter Details, Hospitals, Food Centers, Emergency Contacts	Emergency Resources

4.8 External Interface Requirements

- **User Interface:** Web-based dashboard and mobile application for citizens, government authorities, and rescue teams.
- **Hardware Interface:** Computer, Server, Internet connection.
- **Software Interface:** Database Management System (DBMS), Weather APIs, AI Prediction Module, and Notification Services.
- **Communication Interface:** Secure HTTPS protocol with SMS, Email, and Mobile Push Notification services.

4.9 System Features

- Secure user registration and authentication.
- Real-time disaster monitoring and AI-based risk prediction.
- Location-based disaster alerts and emergency notifications.
- Nearby shelter, hospital, food aid, and first-aid center locator.
- Emergency contact information and government authority support.
- Disaster report generation and historical record management.
- Role-based access control for citizens, administrators, and rescue teams.

5 Identify Assumptions and Constraints

5.1 List the assumptions considered while developing the system.

- Users have smartphones and internet access to receive disaster alerts.
- Real-time weather information is available through trusted weather APIs.
- Users enable GPS/location services for location-based notifications.
- Government authorities and rescue teams regularly update emergency information.
- The system database is maintained and backed up regularly.

5.2 Identify technical, operational, economic, legal, and time-related constraints affecting the project.

Constraint Type	Description
Technical	Requires stable internet connectivity, weather APIs, and reliable server infrastructure.
Operational	Citizens, authorities, and rescue teams should use the system during emergencies.
Economic	Development and maintenance costs should remain within the project budget.
Legal	The system should comply with data privacy and government regulations.
Time	The project should be completed within the specified development schedule.

6 Validate Requirement Completeness and Consistency

6.1 Verify whether all stakeholder requirements have been addressed.

Validation Check	Status
Stakeholder requirements addressed	✓ Yes
Functional requirements complete	✓ Yes
Non-functional requirements complete	✓ Yes
Use cases identified	✓ Yes
Data and interface requirements included	✓ Yes

6.2 Check for ambiguity, redundancy, inconsistency, and missing requirements.

Aspect	Observation
Ambiguity	Requirements are clearly defined and easy to understand.
Redundancy	No duplicate requirements are identified.
Consistency	Functional and non-functional requirements are consistent with the system objectives.
Missing Requirements	Future enhancements may include IoT sensors, offline alerts, and multilingual support.

6.3 Suggest improvements to enhance the quality and completeness of the requirement specification.

- Improve disaster prediction accuracy using advanced AI and Machine Learning models.
- Integrate IoT sensors for real-time environmental monitoring.
- Add multilingual support to help users from different regions.
- Include offline emergency alerts and live rescue tracking for better emergency response.

6.3 SUGGEST IMPROVEMENTS TO ENHANCE THE QUALITY AND COMPLETENESS OF THE REQUIREMENT SPECIFICATION.



Improve disaster prediction accuracy

Improve disaster prediction accuracy using advanced AI and Machine Learning models.



Integrate IoT sensors

Integrate IoT sensors for real-time environmental monitoring.



Add multilingual support

Add multilingual support to help users from different regions.



Include offline alerts and live tracking

Include offline emergency alerts and live rescue tracking for better emergency response.

